

Final Prototype Memo

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Background & Problem Statement:

Our team was approached by Prof. Santana, an engineering professor at Harvey Mudd College, who often organizes outreach events at local elementary public schools to cultivate experimentation and excitement about engineering and science. Drawing from his own research on hydrogels, Prof. Santana is looking for a physical test setup for outreach events that allows elementary students (specifically 4th and 5th graders) to interactively learn the material behavior of hydrogels.

Final Prototype:



Figure 1: Final Functional Prototype

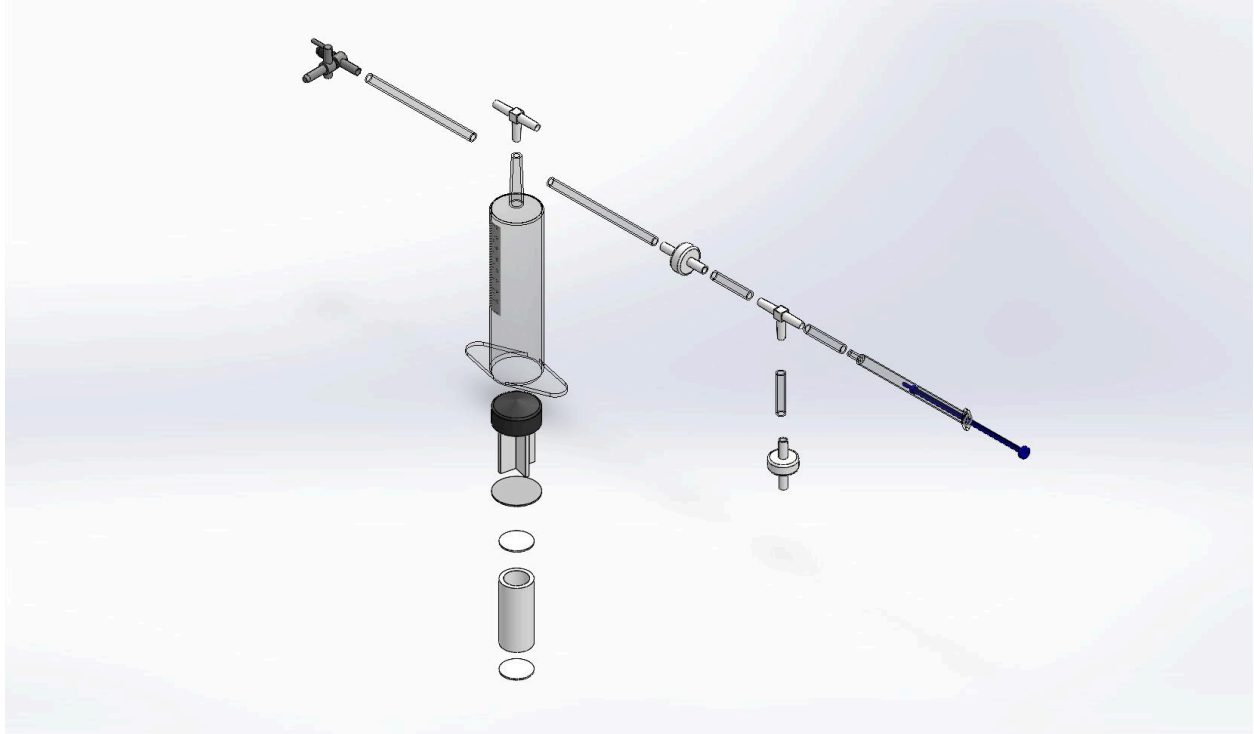


Figure 2: Exploded View of the Final Functional Prototype

Final Functional Design Justification: Objectives and Constraints

Objectives:

	Objectives	Explanation/Reasoning
A	Minimize setup time	Allows for children to setup quickly so they can finish the experimentation in a small amount of time
B	Minimize cleanup time	Allows for the children to cleanup in a small amount of time so they can move on to their next subject or class
C	Maximize durability and reusability	Enable repeated use without loss of performance or damage.
D	Minimize maintenance	Ensure components are easy to clean (ideally dishwasher safe) and quick to replace.
E	Maximize ease of use	Allow children to independently set up, operate, and reset the device with minimal instruction.

F	Minimize difficulty to assemble/make	Allow an adult to easily make multiple units in a short time
G	Maximize visibility	Allow kids to clearly observe the material during experimentation
H	Minimize cost	Allow lower income students/educators to afford this product.

For each objective, we developed the following metrics, collectively organized in Tables 1-8, so that subsequent designs can be objectively evaluated and scored:

Minimize setup time	Type of Scoring: Quantitative (minutes)	
	Description How long it takes to set up and clean up the mechanism and working space	Score
	0 - 10 minutes	4
	10 - 15 minutes	3
	15 - 20 minutes	2
	20 - 30 minutes	1
	30+ minutes	0

Table 1: Metric for minimizing setup time

Minimize cleanup time	Type of Scoring: Quantitative (minutes)	
	Description How long it takes to set up and clean up the mechanism and working space	Score
	0 - 10 minutes	4
	10 - 15 minutes	3
	15 - 20 minutes	2
	20 - 30 minutes	1
	30+ minutes	0

Table 2: Metric for minimizing cleanup time

Maximize durability and reusability	Type of Scoring: Qualitative	
	Description: Measured as how much functional and aesthetic damage the design sustains after 5 uses	Score
	The mechanism sustains no damage after 5 uses	4
	The mechanism sustains some aesthetic damage after 5 uses; no impact on functional use	3
	The mechanism sustains both aesthetic and functional damage after 5 uses. However, it is repairable with little effort and money from the user.	2
	The mechanism sustains both aesthetic and functional damage after 5 uses. It requires significant effort and money to repair.	1
	The mechanism sustains permanent aesthetic and functional damage after 5 uses.	0

Table 3: Metric for maximizing durability and reusability

Minimize maintenance	Type of Scoring: Quantitative	
	Description: Measured by the amount of effort required to keep the mechanism functional	Score
	Maintenance is very easy and is quick to finish	4
	Maintenance is very easy but takes some time	3
	Maintenance is moderately tiring and a little time consuming	2
	Maintenance is very tiring and a little time consuming	1
	Maintenance is very tiring and time consuming	0

Table 4: Metric for minimizing maintenance

Maximize ease of use	Type of Scoring: Qualitative	
	Description: A user-reported metric of how much there were confused while using design	Score
	User was able to use the design correctly without any confusion or strenuous effort	3
	User asked one question or required a small hint to use the design correctly	2
	User asked for multiple questions or required multiple hints to use the design correctly	1
	User required complete intervention by the team while using design	0

Table 5: Metric for minimizing ease of use

Minimize difficulty to assemble/make	Type of Scoring: Qualitative	
	Description: Defined as the amount of time and effort required to create the unit	Score
	Less than 30 minutes	4
	30 - 40 minutes	3
	40 - 50 minutes	2
	60 minutes +	1
	Not Able to Assemble	0

Table 6: Metric for minimizing difficulty to assemble/make

Maximize visibility	Type of Scoring: Qualitative	
	Description: How well the children can see the boba deform	Score
	Can see the boba deformation very clearly	4
	Can see important deformation, but only some of them to a precise measurement	3
	Can see important deformation but not to a very precise measurement	2
	Can see some deformation of the boba but not enough to develop a good understanding of material deformation	1
	Can't see any boba during experimentation	0

Table 7: Metric for maximizing visibility

Minimize Cost of Final Unit	Type of Scoring: Quantitative	
	Description: Bill-of-materials required to make to make one Boba-Buster unit	Score
	Less than \$7/unit	4
	Between \$7/unit and \$10/unit	3
	Between \$10/unit and \$12/unit	2
	Between \$12/unit and \$15/unit	1
	Exceeds \$15+	0

Table 8: Metric for minimizing cost of final unit

Constraints:

Time Constraints	Must conduct the experimental demonstration within a 1 hour time period • Prof. Santana requested that the experiment should be completed within a normal class time period.
Material Constraints	Must use components made from non-toxic and children-safe materials • A big component of working with elementary school kids is their safety. If we do not put any safety constraints on potentially dangerous materials, it causes harm to the children's health. Health always takes priority over learning subject material. Must use off-the-shelf, easily accessible materials • The mechanism should not require instructors to go to specialty stores. ◦ Examples: Target, Home Depot Must be tested on popping boba • This is from Prof. Santana's request as popping boba is cheap, accessible, and food-safe (in the case, students eat it).
Safety Constraint	Must have no sharp edges, pinch points, or high force mechanisms • Similar to our reasoning for the material constraints, the students' health and safety take priority over their learning.
Tool Constraint	No advanced machinery and must be manufactured and assembled with common tools • A local public school will likely not have access to advanced resources.
Budget Constraint	Must not exceed \$15/unit • Prof. Santana ideally prefers that the product be priced around \$5 per unit. However, Prof. Santana will ultimately be taking advantage of the economies of scale and buying a bulk set of material. Hence, we will be using a constraint of \$15/unit throughout the project.

How Our Prototype Meets The Objectives:

All scores for each objective refer to the metric tables defined above.

A: Minimize experiment setup time - 4

The user simply needs to place one popping boba on the stand to conduct one round of the experiment. Prof. Santana's children took about 30 seconds to set-up with verbal instructions. The low set-up time further corresponds to low complexity in the prototype.

B: Minimize cleanup time - 4

The user only has to wipe the inside edges of the 60 mL syringe barrel, the end of the 60 mL syringe plunger, and the ends of the ½ PVC Coupling with a damp paper towel to make sure all the sticky residue is removed before the next use.

Additionally, our entire mechanism is waterproof. A user can wash the entire mechanism under water without impacting its functionality.

C: Maximize durability and reusability - **3**

In testing the prototype across various users, we observed how the rubber gasket of the smaller plunger sustains some physical wear with noticeable fraying on the edges. The fraying makes the pumping require more effort as air can leak and make the main chamber less air tight. However, this is very minor damage and the mechanism is still fully functional, which justifies the rating of **3**.

D: Minimize maintenance - **4**

A strength of this prototype is its high modularity, with parts easily being able to be swapped out. This simplifies maintenance, minimizes downtime, and keeps the experimental setup operational with minimal disruption.

E: Maximize ease of use - **3**

The small syringe delivers only a limited volume of air, requiring children to depress it multiple times to pop the boba, which may lead to shoulder fatigue. However, this is the only task they are required to perform, so the overall ease of use remains high. Furthermore, the syringe's size is essential for maintaining the precision of the data collected. Therefore, we optimized ease of use while adhering to the precision requirements of the experiment.

F: Minimize difficulty to assemble/make - **4**

The user is only required to cut three pieces of tubing and the plunger of the large syringe. The remainder of the assembly involves straightforward application of hot glue and tape following the instructions provided in the process router.

G: Maximize visibility - **4**

The clear 60 mL syringe barrel enables the user to effortlessly see when the boba is compressed and bursts. Moreover, the stand raises the boba, offering a better viewing angle. The millimeter indicators on the syringe's side allow the user to measure the boba's deformation accurately.

H: Minimize cost - **2**

Each unit costs \$11.26 to make. Using this per unit cost and buying parts individually, schools may not be able to afford enough units for the volume of a typical classroom setting (approximately 20 to 30 students). However, if materials are purchased in bulk to support a larger number of units (like for a whole class), it is possible to reduce per unit cost through economies of scale.

How Our Prototype Meets The Constraints:

Time Constraint: Our mechanism must be able to teach 4th and 5th graders within 1 hour. It took our group only 20 minutes to conduct 3 trials of popping the boba and charting data. However, we estimate that 4th and 5th grade students will require around 30 to 40 minutes. This estimate accounts for the reduced attention span, lack of familiarity with experimental procedures, and reduced dexterity in 4th and 5th graders. Thus, the 4th and 5th graders will have enough time to conduct their experiment in the allotted hour, so our mechanism meets the time constraint.

Material Constraints: Our mechanism must meet three material constraints.

- Must use components made from non-toxic and child-safe materials
 - Our mechanism remains non-toxic and safe for children to use because we only use plastic, copper, rubber, tape and hot glue.
- Must use off-the-shelf, easily accessible materials
 - All of our materials are obtainable from Home Depot or Amazon.
- Must be tested on popping boba
 - Data collection during our prototype evaluation (Table 10) shows that our mechanism is able to effectively compress popping boba.
- Students allergic to popping boba or any of the materials in the mechanism should notify their instructor.

Safety Constraint: Our mechanism must have no sharp edges, pinch points, or high force mechanisms. The material is blunt and smooth, removing the possibility of cutting oneself. This can be confirmed through Figure 1 and the testing of our prototype. Additionally, the only pinch points in the mechanism are between the small connectors and tubing.

Tool Constraint: The assembly process must not use advanced machinery and must be performed with common tools. We only used scissors, tape, and hot glue.

Budget Constraint: The mechanism must not exceed \$15/unit. Our current prototype is \$11.26/unit.

Final Functional Design Justification: Material Choices

60 mL Luer-Slip Syringe Barrel, 1 mL Luer-slip Syringe, 60mL Syringe Plunger and 60mL Syringe Plunger End, One-way Check Valve, Three-way Connector:

The syringe components and plastic valves are made from PP and PVDF (variants of plastic), which is waterproof and can be cut and altered easily with basic manufacturing tools. Additionally, it is important for the children to visualize the deformation of the boba, so using a transparent plastic like PP in the syringe maximizes the visualization of the boba deformation.

Vinyl Tubing:

We chose PVC tubing for our mechanism because it is cheap and safe for children to handle. Minimizing cost is a key priority for our client, so we selected a material that offers a good balance of affordability and durability.

Fasteners:

Duct Tape:

- Duct tape is cheap and flexible. It secures the tubing without applying excessive pressure that could cause damage or limit airflow.

Hot Glue:

- Hot glue is cheap, air-tight, and quick to dry.

Single-way T valve:

- The choice of material for the single-way t-valve is not crucial, so we chose the least expensive option available.

½ PVC Coupling (Boba Stand)

- A ½-inch PVC coupling works well as a boba stand because of its stability, dimensions, and strength. The cylindrical form with flat surfaces creates a stable base that holds the boba in place during testing. The coupling endures multiple trials without deforming as well. Furthermore, it is lightweight, affordable, and simple to manage, making it ideal for prototype applications in educational environments. In addition, the coupling can be easily cut, sanded, or connected to other parts, enhancing alignment and stability within the system

Ends of PVC Coupling, End of 60 mL Syringe Plunger

- The flat ends of the PVC coupling and the end of the 60 mL syringe plunger are made from plastic. It is lightweight, durable, and easy to shape into the precise geometry needed for our PVC coupling. It also resists moisture and deformation, ensuring that the ends remain firm even after continuous testing with boba. Additionally, plastic is inexpensive and widely available, making it a practical choice for repeated classroom or outreach use.

Prototype Functionality:

Functions

Measures and displays load acting on boba	The product should measure and provide a clear indication of the magnitude of load acting on boba.
Applies compressive load to boba	The product should be able to apply compressive load to the boba.
Measures the deformation of boba	The product should measure some physical aspect of the boba as it gets deformed.
Stabilizes against external forces	The product should withstand forces applied by a child using the design.

How The Prototype Has All These Functions:

Measures and displays load acting on boba:

Our mechanism displays the deformation to the user because the syringe is transparent. The 1 mL syringe has printed graduation marks to show much air is pumped into the system, which corresponds to the load applied on the boba.

Applies compressive load to boba:

The small syringe pumps air into the 60 mL syringe barrel, enabling the user to apply a vertical force on the boba using air pressure.

Measures the deformation of boba:

A millimeter-increment measuring strip is taped on the side of the 60 mL syringe. Deformation of the boba is quantified by measuring the height of the boba.

Stabilizes against external forces:

The mechanism stays intact even when it is dropped on the table or pumped aggressively. While the mechanism requires an additional user to keep the 60 mL syringe barrel stable, this contributes to greater consistency. The user can read and record measurements of changes in height of the boba leading to more consistent experimental results.

Question Our Prototype Answers:

Using this final prototype, we aimed to answer the important question:

Is our final prototype functionally consistent and easy for instructors to learn how to use?

While we ideally would have liked to test the final prototype with children, our progress in the project and constraints with the project timeline did not allow for this. We instead had other college students use the prototype to emulate an instructor's first experience with the prototype.

Evaluation Plan

To evaluate our prototype and answer our prototyping question above, our team used the following evaluation procedure and metrics.

Evaluating consistency:

1. Place the assembled prototype on top of a single piece of popping boba rested on our stand (as seen in Figure 3), with the 60 mL syringe's base flat on an even surface.
 - a. Record the reference height of the boba by using the ruler on the side of the big syringe.

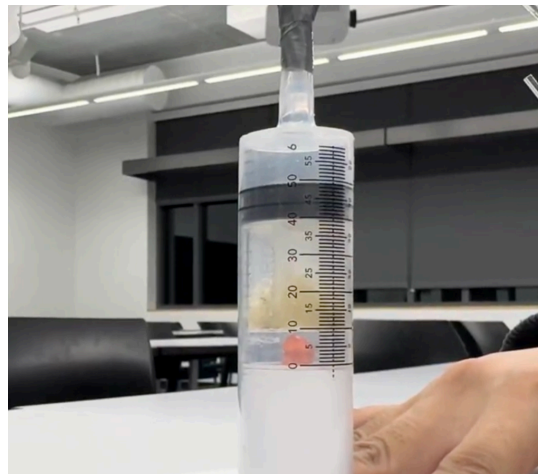


Figure 3: The prototype loaded with boba.

2. Use the small syringe to slowly pump air into the big syringe.
 - a. Recordings of the air volume should not be taken until the 60 mL plunger end is visibly touching the top of the boba
3. Once the plunger end is touching the top of the boba, begin taking experimental data.
 - a. After every pump of the small syringe, record the amount of air volume and any change in height of the boba using the ruler sticker on the 60 mL plunger barrel.

4. Note when the boba pops as the final data point.
5. Use paper towels to remove the remaining film of popping boba and liquid residue from the base of the 60 mL plunger and the stand.
6. Release air pressure by opening the single-way t-valve on the side of the mechanism.
7. Repeat steps 1 through 5 two additional times for 3 trials.
8. Calculate the relative standard deviation to determine consistency.

Description - Consistency as determined by relative standard deviation ($s/\bar{x} * 100\%$)	Score
Excellent Consistency (<2%)	5
Good Consistency (2% - 5%)	4
Fair Consistency (5% - 10%)	3
Poor Consistency (10% - 15%)	2
No consistency (>15%)	1

Table 9: Metric for evaluation consistency

Evaluating Ease of Use:

1. Read out the instructions from the Boba Busters user manual to the test subject, and have the test subject use the mechanism.
2. Ask the test subject to rate the mechanism using Table 5 (Maximize ease of use), and collect their ratings.
3. Repeat steps 1-2 four additional times with 4 different subjects for a total of 5 trials
4. Calculate the average for all the trials

Evaluation Results:

Evaluating consistency:

Table 10 organizes the results of the three trials, with the calculated average ($\bar{x}$) and the relative standard deviation ($s/\bar{x} * 100\%$ where s is the sample standard deviation). From Table 10, we notice that most of the trials required a similar amount of air and resulted in an average air volume of 12.5mL .

Trial #	1	2	3	Average	Relative Std
Air volume to pop boba (mL)	12.5	12	13	12.5	4%

Table 10: Numbers of trials and required air to pop boba

Our results fell within the "good consistency" range shown in Table 9, with a relative standard deviation of 4%, indicating that the device consistently produces comparable air pressure throughout trials. Variations in the boba themselves, such as changes in moisture content, geometry, or external forces are most likely responsible for the slight variations in detected air volume. These results corroborate the mechanism's low-maintenance and high-durability features by showing that it operates consistently after repeated use.

Height (mm) vs. Air Volume (mL)

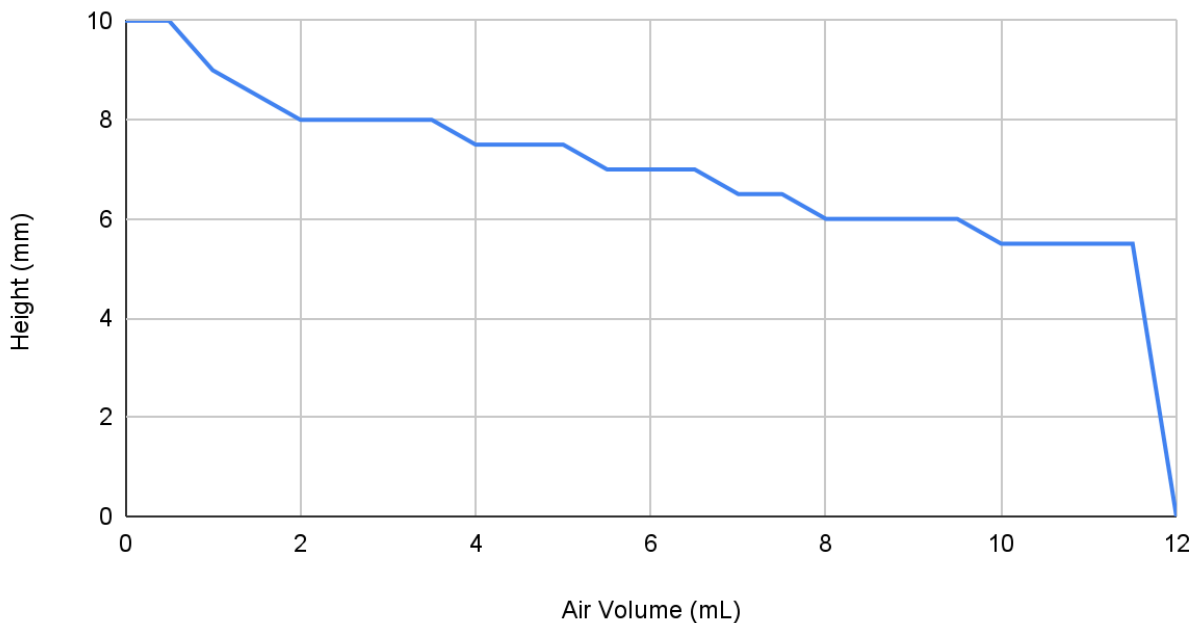


Figure 4: Example of graph produced from plotting height (mm) vs. air volume (mL)

Additionally, Figure 4 demonstrates an example of a graph a student may produce from their recorded data. The figure plots height against air volume and the analyzable data shows that

the mechanism achieves its functions: it measures and displays load acting on boba, applies a compressive load to boba, and measures the resulting deformation.

Evaluating ease of use:

	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Average	Standard Deviation
Rating	3	3	3	3	2	2.8	0.4

Table 11: Scores that rate our mechanisms ease of use

Table 11 illustrates that the majority of users gave the mechanism a 3 for ease of use, while only subject 5 gave it a 2. Subject 5 expressed concern about the single-way t-valve, noting that understanding which direction to turn the valve to restrict or open airflow was confusing. While instructors might resolve this through trial and error, such confusion might be even more noticeable and disruptive among fourth and fifth graders. Therefore, the mechanism should have a visible indicator which makes it clear which way to adjust the valve in order to consistently receive a score of 3.

It is important to note one significant limitation in this evaluation plan. As mentioned previously, we were unable to test the prototype with actual children due to time constraints in the project, so extrapolation of our results to fourth and fifth-graders is not fully accurate. That said, testing with college students still provided valuable insight such as the problem with usability in the single-way t-valve.

Impact of Design & Next Steps:

Our finished prototype significantly advances Prof. Santana's outreach objectives by offering fourth and fifth-graders an inexpensive, safe, and consistently effective tool that illustrates hydrogel behavior in an interactive manner. First, our evaluation results reveal that the prototype achieves strong functional consistency, with a relative standard variation of 4%, indicating that students would reliably observe the same qualitative conclusion (a gradual compression followed by a clear “pop”). This reliability is vital for teaching scientific repeatability and allowing instructors to consistently demonstrate material behavior.

Second, the prototype's cost (\$11.26/unit) and use of widely accessible materials directly satisfy the client's requirement to scale this activity across several schools and outreach events. Because the mechanism can be made with basic equipment, replicated rapidly, and repaired using modular components, it reduces the preparatory and maintenance burden on instructors.

Third, the device boosts the educational value of the outreach activity by offering primary pupils a clear visual and physical relationship to hydrogel deformation. The transparent components and millimeter-scale measurement sticker let students quantify a physical phenomenon, teaching children about experimental design and measuring material deformation. The design also naturally fosters collaboration where one student pumps the mechanism and measures air volume while another stabilizes it and measures deformation of the boba, making the activity more interacting and engaging.

The main limitation of our current design is that there is significant friction between the plunger and syringe. This occasionally causes the 60 mL plunger to get stuck in the barrel and increases the effort required to move it down. When considering young target users and long term use, this should be reduced. Future prototypes can use a silicone-based lubricant on the rubber part of the plunger to ensure smoother movement. Additionally, based on the feedback provided during prototype evaluation, future versions should include a label or sticker on the single-way t-valve that informs the user the direction to open/close it. Finally, the cost of the prototype can be reduced by purchasing the materials in bulk.